

A Focused Human Body Model for Accurate Anthropometric Measurements Extraction

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Abstract

3D anthropometric measurements have a variety of applications in industrial design and architecture (e.g. vehicle seating and cockpits), Clothing (e.g. military uniforms), Ergonomics (e.g. seating) and Medicine (e.g. nutrition and diabetes) etc. Therefore, there is a need for systems that can accurately extract human body measurements. Current methods estimate human body measurements from 3D scans, incurring a heavy data collection burden. Moreover, slight variations in camera angle, distance, and body postures may significantly affect measurement results. In response to these challenges, this paper introduces a focused human body model for accurately extracting anthropometric measurements. Concretely, we design a Bypass Network via CNN and ResNet architectures, which augments the frozen backbone SMPLer-X with additional feature extraction capabilities. Furthermore, to boost model effectiveness, we propose a dynamical loss function that automatically recalibrates the weights to make the network focus on targeted anthropometric parts. In addition, we construct a multimodal body measurement benchmark dataset consisting of depth, point clouds, meshes and corresponding body measurements to support model evaluation and future anthropometric measurement research. Extensive experiments on both open-source and proposed human body datasets demonstrate the superiority of our approach over existing counterparts, including current mainstream commercial body measurement software.

1. Introduction

Nowadays, more and more consumers go shopping online for perfectly-fitted apparel, driving the demand for mass

personalization and custom-fitted clothing. Accurate body measurements are essential to meet this demand. The growing popularity of reconstructing 3D human body shapes from images has expanded the applications of anthropometric measurements across various fields, including medicine science [25], industrial design [13, 26, 44], fashion industry (clothing design) [8], ergonomics [28], fitness [35], and entertainment [5, 10, 12].

Most existing anthropometric measurement methods can be roughly divided into two types: one-stage framework [2, 22, 33], and two-stage framework [14, 36, 41]. The former calculates body measurements directly from input images or landmarks in an end-to-end manner. However, image-based methods often struggle with scale ambiguity, making them sensitive to noise and hard to obtain accurate measurements. In contrast, two-stage methods overcome these issues by taking body measurements from a deformed template model (e.g. SMPL [21] and SMPL-X [27]), which corresponds to a complete reconstruction of input data. Therefore, they rely heavily on the accuracy of the reconstructed parametric human body model.

Some other methods [22, 29, 40] require capturing point clouds as input and extracting measurement factors from 3D scans. As point cloud representations rely on accurate 3D data acquired via expensive 3D sensors, consequently, several studies turn to utilize statistical models [6], linear regression [2], and convolutional neural networks [14, 36] for body shape measurements. However, these methods have limitations in being generalized to complex and variable scenarios. Factors such as gender, camera angle, distance, and challenging postures impede them to obtain accurate measurements. Recently, Vision Transformer (ViT) models have achieved great success in several 2D and 3D tasks. Although researchers have tried to employ deep learning techniques for anthropometric measurements, there is no ViT-based model for extracting human body measurements

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using two-stage methods. This is partly due to the challenges in gathering paired 3D data with corresponding labeled measurement values and the difficulty of training large-scale models with 3D data.

To overcome the above problems, this paper introduces a large-scale body reconstruction model designed specifically to generate accurate measurement results in complex scenarios. We focus on optimizing a standard anthropometric measurement pipeline, which involves (1) reconstructing the SMPL model parameters to obtain human body meshes and (2) estimating the body measurements from the recovered meshes. They are referred to as two problems: Human Mesh Recovery (HMR) and Human Body Measurements Estimation (HBME), respectively.

Concretely, inspired by the success of large-scale human body models [3] and the wide adoption of template-based approaches, we adopt the encoder-decoder ViT architecture as the backbone model to obtain reconstructed SMPL body parameters. Furthermore, we propose a bypass network to guide the frozen SMPL-X Model to focus on measurement parts. This bypass network, incorporating residual connections, enhances the feature extraction capability of the SMPLer-X model. Additionally, we combine the bypass network with a dynamic loss function to recalibrate the anthropometric and non-anthropometric regions, thereby generating accurate body measurements.

As image-based methods raise privacy concerns related to facial features, resulting in a scarcity of publicly available circumference datasets. We construct a benchmark dataset, named Fashion-body, which includes human depth data, point clouds, meshes, and corresponding anthropometric measurements. Our dataset consists of examples collected from diverse angles, lighting conditions, and backgrounds. In addition, we also validate our method on point cloud data acquired from Mojave Flash Lidar Sensor to effectively prevent the leakage of users' private facial information [23, 45].

The major contributions of this paper are as follows:

- We propose a novel method for extracting anthropometric measurements using a ViT-based SMPLer-X human model from images or point clouds captured with a high-end Mojave Flash Lidar Sensor.
- We introduce a bypass network with a dynamic loss function to the foundation model SMPLer-X. By making the foundation model focus on location-specific reconstruction, both reconstruction performance and the accuracy of anthropometric measurements are significantly improved.
- To support the training and evaluation of the proposed model, we construct a new dataset Fashion-body, which can also serve as a benchmark for future research in anthropometric measurements.

2. Related Work

Human Body and Measurements Datasets The Shape Completion and Animation of People (SCAPE) model [1] significantly advanced the field of human shape estimation by providing a method to build realistic 3D human body meshes in various shapes and poses. While datasets like CAESAR [30] offer extensive 3D scans and body measurements, they lack real images, necessitating simulation from scans using virtual cameras. Recently, Yan et al. [43] published the BodyFit dataset, featuring over 4K body scans with computed body measurements and simulated silhouettes, along with a collection of photographs and tape measurements from 194 subjects. Additionally, BodyM [31] is notable as the first large-scale dataset that pairs body measurements with silhouettes obtained through semantic segmentation of real photographs. However, a comprehensive dataset containing RGB images, point clouds, meshes, and corresponding ground truth measurement values is currently unavailable. In this work, we leverage point clouds, distance fields, and meshes with corresponding measurement values as our Fashion-body dataset.

Body measurements can be extracted from point clouds, template meshes, or features from 3D/2D data. Traditional tools and 3D scans typically provide higher accuracy in obtaining body measurements directly, but they are cumbersome and relatively expensive. To address these issues, researchers have developed a series of fully automatic methods to reduce both the time and cost involved in the measurement extraction process.

Anthropometric Measurements Extraction Anthropometric measurements can be extracted from images, point clouds, template mesh, or features. Traditional tools and 3D scans typically provide higher accuracy in obtaining body measurements directly, but they are cumbersome and relatively expensive. To address these issues, researchers have developed a series of fully automatic methods to reduce both the time and costs required for the measurement extraction process. The existing approaches to anthropometric measurement extraction can be broadly divided into two categories: landmark-based [22], [4] and template-based approaches [42]. Landmark-based approaches involve scanning the human body, body segmentation, landmark detection, and then extracting measurements based on these landmarks. For example, chest measurements are landmarks used to measure the bust point distance. While landmark-based methods can efficiently calculate measurements from identified landmarks, they are sensitive to noise and missing data in the input scan. In contrast, template-based methods calculate measurements on the deformed human template that corresponds to the complete reconstruction human body. These methods overcome the noise and

missing data by fitting a human template body to the input scan. However, the major challenge of existing template-based methods has limitations in generalizing to different scene categories. Additionally, the reconstruction quality lacks reliability under extreme viewing angles, changing backgrounds, and different shooting distances, making accurate measurement across various scenarios challenging. To address this, we combine Vision Transformer (ViT) with the template-based approach to harness the effectiveness and power of large-scale models.

Human Mesh Reconstruction The literature of recovering 3D human representations from RGB images is extensive [37, 39] with techniques falling into two broad categories. Parametric methods [1, 21] characterize the human body model in terms of a parametric representation. Non-parametric methods directly regress a 3D body representation from images using convolutional neural networks [17], transformers [20], intermediate representations [24] or implicit functions [7, 32]. Recently, Lin et al [19] proposed the first ViT-based backbone [9] to resolve the issues in previous approaches. This provides a promising and concise way to leverage the scaling-up model in two-stage body measurements. However, there remains a scarcity of benchmark datasets for comparing body measurements, and few researchers are exploring the integration of additional data for generalizable and accurate body measurement results.

3. Method

3.1. Overview

Our objective is to acquire a mapping function $f(I) = (\theta, \beta, \pi)$ for reconstructing an individual depicted in the image. This involves predicting his/her 3D pose and shape parameters (θ and β , respectively) based on the provided image, where π denotes the camera translation. The function $M(\theta, \beta)$ then maps shape and pose parameters to vertices. As illustrated in Fig. 1, we employ the Vision Transformer (ViT) architecture to acquire a mapping function $f(I) = (\theta, \beta, \pi)$. Then, we utilize a Bypass Network to provide additional supervision to generate accurate results on measurement parts. Observing that the original loss tends to optimize global key points rather than the local measurement points we focus on, we introduce a dynamical loss function that constantly re-weight the parameters between the measurement part and the non-measurement part. On the other hand, the measurement set constitutes our measurement modules, which are pre-defined based on ISO 8559 standard [18]. Finally, we obtain the body measurements by inputting the reconstructed vertices into an anthropometric module.

3.2. SMPL-X Model

In this paper, SMPL-X [27] is used to jointly model the human body, face, and hand on top of SMPL. Initially, there are 75 rotational parameters for the global rotation and body, eyes, jaw poses, 24 low-dimensional PCA coefficients or 90 rotational parameters for hand poses, and 10 for the body shape and 10 for the facial expressions. The joint regressor $\mathcal{F}$ is utilized to obtain 3D key points from the above parameters through a transformation function $\mathcal{R}_\theta$ along the kinematic tree.

3.3. Focused Human Body Model

Our goal is to develop a scalable and efficient model, pre-trained on large datasets, to create body measurement vertices applicable across diverse scenarios, laying the foundation for future studies involving larger models in this field.

Human body model architecture. As illustrated in Fig. 1, we employ a modified Vision Transformer(ViT) [9] for extracting image tokens. The standard transformer decoder [38] consists of six layers, each incorporating multi-head self-attention, multi-head cross-attention, and feed-forward blocks with layer normalization. The decoder has a 2048 hidden dimension, 8 (64-dim) cross-attention heads, and a hidden dimension of 1024 in the feed-forward MLP block. It operates on a single learnable 2048-dimensional SMPL query token as input and cross-attends to the output image tokens. Before entering the backbone, the image undergoes cropping by a whole-body bounding box and resizing to 512×384 . It is then tokenized into 32×24 patches with a patch size of 16, followed by patch embedding and positional encoding to obtain image tokens I_{token} . Additional learnable tokens are concatenated with the SMPL query token and processed utilizing ViT. A positional module is employed to predict 3D key points. Finally, a linear regressor module read out 3D key points from the transformer decoder, providing pose θ , shape β , and camera π .

Focusing network. Our focusing network employs bypass network to extract various guidance features, preserving the pre-trained features for consistent performance and promoting efficient feature extraction. As depicted in Fig. 1, we employ a lightweight module based on CNN and ResNet architectures, serving as a plug-and-play bypass network. This network enhances the feature extraction capabilities of the backbone SMPLer-X model, preventing the discard of raw image information during data transfer. In addition, we utilize SirenReLU [34] as an activation function to efficiently extract features from the image, facilitating precise human reconstruction. This activation function is designed to represent and learn high-frequency functions, especially those complex patterns that are difficult to represent in conventional neural networks.

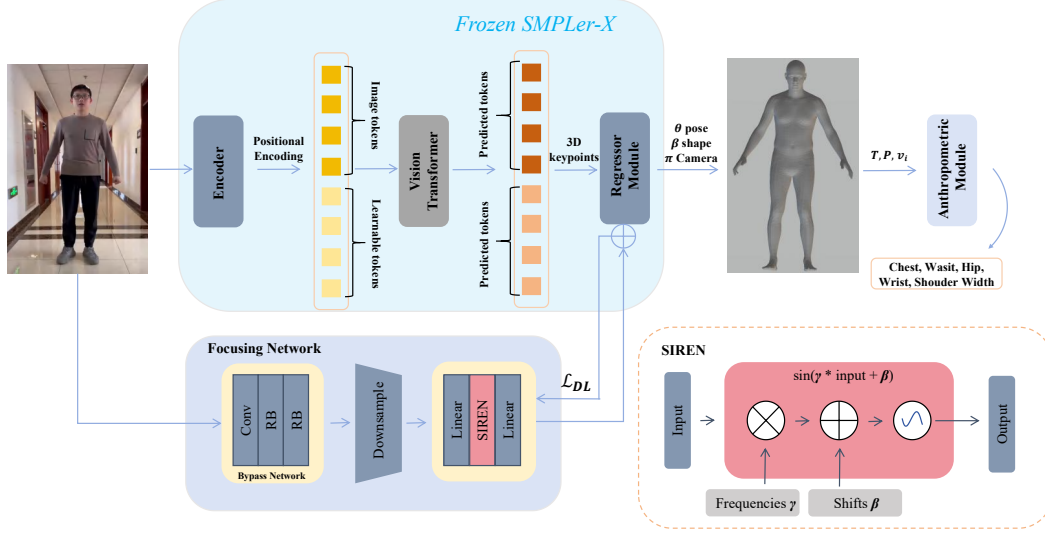


Figure 1. **The architecture of our method.** Our human body model is based on a frozen SMPLer-X, while a trainable Bypass Network constrains the backbone focusing on the measurement parts. Reconstruction results are constantly refined by dynamic loss $\mathcal{L}_{DL}$. The anthropometric module outputs final body measurements which are calculated based on the ISO 8559 standard.

3.4. Human Body Measurement

Our anthropometric module is based on the SMPL model, which is a parameterized body model derived from numerous high-quality 3D scans representing diverse individuals in varying shapes and poses. This model decomposes body deformations into 72 pose and 10 shape parameters. Initially, we standardize pose and shape parameters for each gender to create an average-shaped mesh in an A-pose. We define an accurate mesh range of measurements on the template of the SMLP model. Then we obtain high-quality mesh and bounding boxes from Focused Human Body Model and approximate the subject height as the height of the bounding box.

Subsequently, we introduce a few notations to describe our anthropometric mathematically. Given an SMPL surface template T , we define a measurement plane $P_i (i \in 1, 2, \dots, k)$ which is a user-defined set of k body measurements, including chest circumference, waist circumference, hip circumference, wrist circumference, and shoulder width, according to the ISO 8559 standard which describes the location of measurements used for the creation of physical and digital anthropometric databases. Fig. 3 visualize the relatively exact anthropometric measurements selected from this standard. And the measurement set $\mathcal{M}_i$ represents the intersection of the surface template T with a measurement plane P_i . Note that a measurement set $\mathcal{M}_i$ is the combination of adjacent measurement vertex $\mathbf{v}_i^{(m)}$ and $\mathbf{v}_i^{(n)}$ (Examples $\mathbf{v}_i^{(1)}$ and $\mathbf{v}_i^{(2)}$ are illustrated in Fig. 2. Finally, we define the mode of measurement set $|\mathcal{M}_i|$ as anthropomet-

ric measurement value:

$$|\mathcal{M}_i| = |\mathbf{v}_1^{(i)} - \mathbf{v}_2^{(i)}| \oplus |\mathbf{v}_2^{(i)} - \mathbf{v}_3^{(i)}| \oplus \dots \oplus |\mathbf{v}_m^{(i)} - \mathbf{v}_n^{(i)}| \quad (1)$$

where $\oplus$ is the row-wise concatenation operator and the Manhattan distance is utilized to calculate adjacent measurement vertex $\mathbf{v}$ which effectively expedites computation. According to the definition above, our measurement module outputs the $|\mathcal{M}_i|$ as anthropometric result.

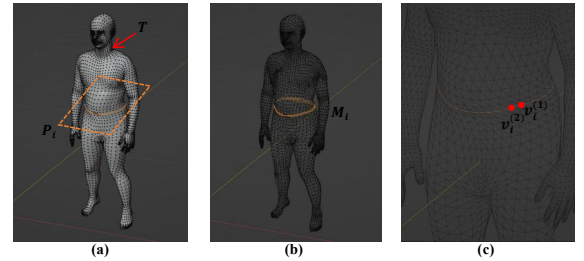


Figure 2. The descriptions of the measurement plane P_i with surface template T and their intersection $\mathcal{M}_i$. Both $\mathbf{v}_i^{(1)}$ and $\mathbf{v}_i^{(2)}$ are the adjacent measurement vertexes belonging to the measurement part $\mathcal{M}_i$.

3.5. Loss Function

We propose a dynamic loss function specially developed to train our architecture and boost the performance of body measurements. our total loss function is a weighted sum of $\mathcal{L}_{\text{coord}}$ and $\mathcal{L}_{\text{para}}$ which are defined in equation (1) and equation (2), respectively.

We introduce a dynamic weight parameter, denoted as

α , associated with the proportion of the measurement part and non-measurement part. The coordination loss $\mathcal{L}_{\text{coord}}$ is defined as follow:

$$\mathcal{L}_{\text{coord}} = (1 - \alpha) \left(\sum_{\mathbf{v}_i \in \mathcal{M}} w_i \|\hat{\mathbf{v}}_i - \mathbf{v}_i\|_2^2 \right) + \alpha \left(\sum_{\mathbf{v}_i^o \in T/\mathcal{M}} w_i \|\hat{\mathbf{v}}_i^o - \mathbf{v}_i^o\|_2^2 \right), \quad (2)$$

where $\hat{\mathbf{v}}_i^o$ and $\mathbf{v}_i^o$ are the i -th predicted and ground truth 3D vertex outside the measurement parts and $\hat{\mathbf{v}}_i$ and $\mathbf{v}_i$ are the i -th predicted and ground truth 3D vertex locations on the measurement parts. $T/\mathcal{M}$ is the vertex on template T outside the measurement set $\mathcal{M}$. The weight w_i for vertex v_i is proportional to the average area of the mesh triangles connected to vertex $\hat{v}_i$.

Whenever we have access to the GT SMPL pose parameters θ^* and shape parameters β^* , a $\mathcal{L}_{\text{para}}$ loss is bootstrapped to the model predictions.

$$\mathcal{L}_{\text{para}} = \|\theta - \theta^*\|_2^2 + \|\beta - \beta^*\|_2^2 \quad (3)$$

Where θ and β are the predicted pose and shape parameters.

Finally, the total dynamic loss of our scaling-up human body model is as follows:

$$\mathcal{L}_{\text{DL}} = \lambda_{\text{coord}} \mathcal{L}_{\text{coord}} + \lambda_{\text{para}} \mathcal{L}_{\text{para}}. \quad (4)$$

4. Experiments

In this section, we conduct comprehensive experiments on the open-source body measurement dataset, proposed Fashion-body dataset, and Lidar data to validate the robustness and effectiveness of our method. We showcase qualitative results of reconstructed body shape and quantitative results of measurement values. Ablation studies are discussed to evaluate the impact of tuning loss and performance when integrating the proposed bypass network.

4.1. Datasets

Open-source body measurement dataset. To evaluate and compare the proposed methods with the state-of-the-art body measurement baselines, we carried out experiments on Body Measurements Datasets [16], including the front and side images of 391 males and 324 females with corresponding GT measurements.

Fashion-body dataset. Previous body measurement datasets lack the details and diversity of real body shapes, such as distance, point clouds, and meshes. Our Fashion-body dataset fills this gap, being the first public dataset to include not only 30 paired front and side images but also distance, point clouds, and meshes. An example is illustrated in Fig. 5. Each subject was measured by a skilled anthropologist, providing chest, waist, hip, wrist, and shoulder

width as the GT measurement values for our experiments. RGB images were captured in an indoor setup, with subjects standing in A-pose and capture distance varied from 2.5-3.5 meters. Note that we never require subjects to wear tight-fitting clothing. We also utilize a high-performance commercial Mojave Sensor for scanning subjects aged between 18 and 24 years old. The sensor outputs amplitude and distance files, which can be converted into point clouds. That allows us to conduct tests on LiDAR data.

4.2. Baselines and Metrics

We compare our method with several state-of-the-art body measurement baseline methods. The methods of comparison are divided into three categories: The first one are End-to-end methods, e.g., Linear Regression [2], NeuralAnthro [36]. Then, self-defined baselines, denoted as **HMR-BMViT** and **4D-BMViT**, are proposed to validate the effectiveness of our proposed pipeline. **HMR-BMViT** utilize the backbone of HMR [15] to reconstruct full 3D mesh in an end-to-end manner. **4D-BMViT** is a fully updated version of HMR, which adopts an end-to-end transformer architecture [11] as the backbone to reconstruct human body mesh. Finally, we select 3 existing commercial 3D body measurement mobile applications and set consistent input forms for a fair comparison with our method. **Verifit**¹ is a 3D body scanning app that allows user to share measurements with Verifit-powered brands and receive accurate size recommendations or custom-fit products. **YourFit** is an AI-powered 3D body scanning solution from 3DLook² company, targeted to precise sizing, virtual fitting, and health and fitness data tracking.

To quantitatively evaluate the measurement results, we utilize the following standard metrics: Mean absolute error (MAE), Relative percentage error (RPE), and Mean Absolute Difference (AMAD). All the evaluation metrics report the mean of these values and measurement values are measured in centimeters (cm).

4.3. Implementation Details

The training of our backbone human body model is conducted on 8 GeForce RTX 4090 GPUs, with a total batch size of 512 (256 for ViT-Huge) for 10 epochs. Specifically, the lightweight focusing network only takes 20 min to train 20 epochs. We use Adam optimizer with cosine annealing for both training and fine-tuning. The learning rate for training is 1×10^{-5} with the minimum learning rate set to 1×10^{-6} , while the learning rate for finetuning is 1×10^{-5} with the minimum learning rate set to 5×10^{-7} . The input image is first patchified into input tokens and passed through the transformer to get output tokens. The output tokens are then passed to the transformer decoder. We use

¹<https://www.verifit.com/>

²<https://3dlook.ai/>

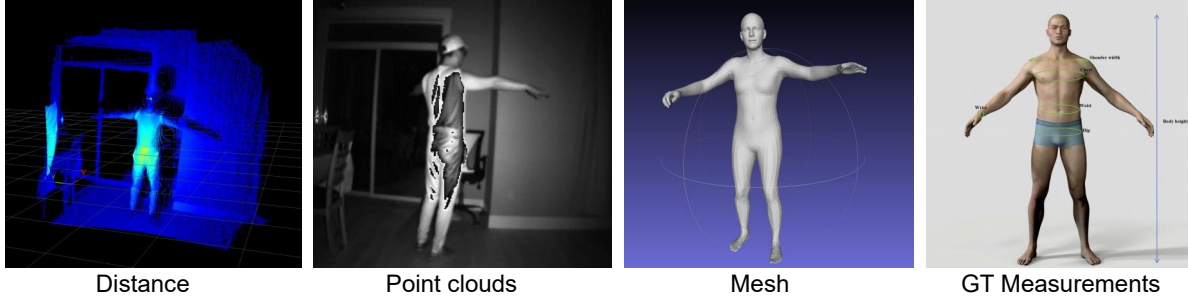


Figure 3. Overview of our proposed dataset. Each object consists of point cloud data, distance from camera, SMPL mesh, and corresponding ground truth body measurements. The samples leverage the Open3D library for visualizing the point cloud and OpenCV for displaying the distance and amplitude images.

a ViT-H/16, the “Huge” variant with 16×16 input patch size. It has 50 transformer layers, takes a 256×192 sized image as input, and outputs 16×12 image tokens, each of dimension 1280. During training, we set λ_{coord} and λ_{para} to 0.2 and 0.1. The parameter α is initialized to zero and gradually increased throughout the iterations.



Figure 4. Reconstruction results of front and side on body measurement dataset and our Fashion-body dataset.

4.4. Qualitative and Quantitative Results

Reconstruction results from images. We present three examples of front-view reconstruction from the body measurement dataset in the 1st row of Fig. 4, and three examples of side-view reconstruction from our Fashion-body dataset in the 2nd row. The subject reconstructs of body measurement showcase good results even under extreme camera views and varying camera distances, facilitating subsequent anthropometric measurements extraction. Furthermore, our method achieves accurate reconstructions for side-view subjects in our Fashion-body dataset, even in

scenes with heavy clothing.

Measurement results from images. Here, we evaluate four measurement parts and five body parts on body measurement dataset and our Fashion-body dataset, each with a single image as input. The corresponding MAE results are shown in Tab. 1. Linear regression [2] is a simple linear model that estimates body measurements based on height and weight. NeuralAnthro [36] employs a convolutional neural network and four additional body dimensions as supervision signals for body measurement estimation. Linear regression achieves MAEs of approximately 9.74 centimeters on the body measurement dataset and 6.23 centimeters on our Fashion-body dataset. NeuralAnthro performs worse, especially in measuring hip circumference. On the other hand, HMR-BMViT achieves lower MAE errors on both datasets with similar performance observed in 4D-BMViT. This improvement can be attributed to the adoption of transformer architecture, which outperforms traditional DNN and CNN networks. More importantly, our method surpasses both HMR-BMViT and 4D-BMViT across all body measurements, due to leveraging the scalability of the focusing network to ensure reliable measurement values. Notably, the deviations in the overall measurements of the baseline methods arise from their inadequate optimization in reconstructing the measurement parts.

Measurement results from point clouds. We obtain the point clouds from the output amplitude and distance of the Mojave Sensor (For the detailed process please refer to appendix). The obtained point clouds are converted into meshes using a self-supervised method [46]. Then, we utilize our anthropometric module — the Point2Mesh baseline, to calculate measurements from the reconstructed 3D body. Fig. 5 demonstrate the qualitative reconstructions for different poses with 360° rotation. For a fair comparison, we re-train all models on the same validation dataset. We present MAE results of HMR-BMViT, 4D-BMViT, and our proposed Point2Mesh in Tab. 2 to demonstrate the quantitative performance of our-BMViT on lidar data. Our-BMViT obtains superior performance across all measurements with a small MAE error in chest, waist, hip, wrist, and shoulder

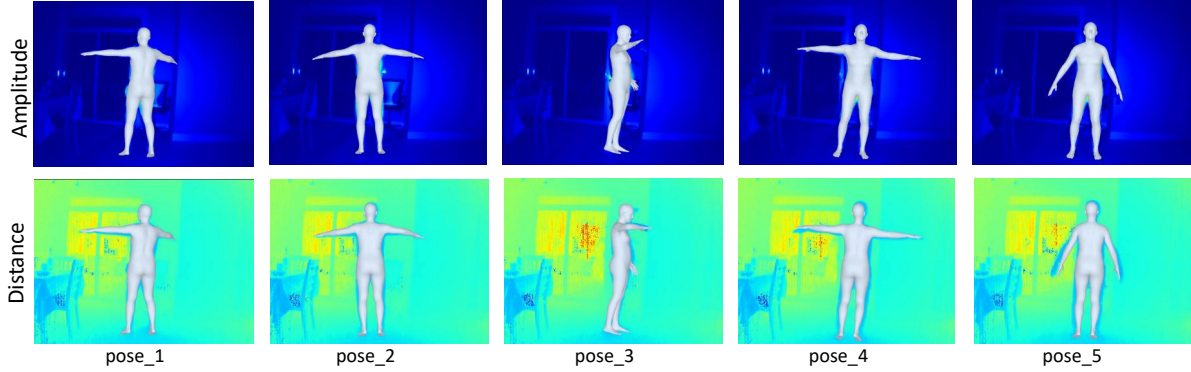


Figure 5. Qualitative body reconstruction results with different poses from amplitude and distance of our Fashion-body dataset.

Method	BODY MEASUREMENTS DATASET					OUR FASHION-BODY DATASET				
	Chest	Waist	Hip	Shoulder Width		Chest	Waist	Hip	Wrist	Shoulder Width
<i>Linear Regression</i>	5.93	7.43	15.98	9.62		4.24	5.52	8.67	2.46	10.26
<i>NeuralAnthro</i>	11.84	12.28	17.7	4.46		10.98	11.89	20.04	1.71	5.15
<i>HMR-BMViT</i>	6.80	10.21	9.29	3.22		6.23	4.88	7.76	0.72	3.88
<i>4D-BMViT</i>	5.19	14.53	13.54	2.37		6.92	11.16	13.7	0.98	3.96
<i>Our-BMViT</i>	3.29	6.73	8.27	2.29		3.07	3.91	6.95	0.41	2.03

Table 1. Mean Absolute Error (MAE) results on body measurements dataset and Fashion-body dataset. We report MAE results of chest size, waist size, and hip size between the predicted circumference and the GT circumference.

width. This is attributed to the reconstructions being well-aligned with the raw amplitude and distance.

Method	LIDAR DATA				
	Chest	Waist	Hip	Wrist	Shoulder Width
<i>HMR-BMViT</i>	4.12	8.70	7.74	1.47	4.67
<i>4D-BMViT</i>	6.25	5.80	5.80	0.74	3.91
<i>Point2Mesh</i>	5.03	3.44	6.40	0.59	1.73
<i>Our-BMViT</i>	3.07	3.41	5.71	0.41	1.53

Table 2. Measurement results (MAE in cm.) on lidar data.

Comparison with mobile applications. To compare our proposed method with mainstream commercial body measurement applications, we conducted experiments on six subjects (three males and three females) utilizing an iPhone 12 to capture the input data. We report the mean absolute error (MAE) for chest circumference, waist circumference, hip circumference, and wrist circumference in Tab. 3. Our method presents slightly better results compared to the commercially available anthropometric software Verifit. While YouFit predicts measurements directly from a 3D scan in the standing pose, it achieves an average MAE of 5.1 centimeters for chest measurements, compared to 3.32 centimeters with our method. Although Verifit outperforms in hip measurements, it exhibits a large deviation in waist measurements. Overall, our method demonstrates superior performance across all dimensions of body measurements. It demonstrates significant improvements in chest, waist, and wrist measurements, outperforming both 3DLook and Verifit in our dataset.

Method	FASHION-BODY DATASET			
	Chest	Waist	Hip	Wrist
<i>3DLook</i>	5.21	4.29	5.62	0.56
<i>Verifit</i>	3.63	8.30	1.19	0.33
<i>Our-BMViT</i>	3.32	3.53	2.64	0.11

Table 3. Average MAE (cm.) comparison with commercial body measurement applications on our Fashion-body dataset.

Method	OUR FASHION-BODY DATASET				
	Chest	Waist	Hip	Wrist	Shoulder Width
<i>SMPLer-X</i>	6.34	9.84	11.05	1.13	4.57
<i>Ours w/o $\mathcal{L}_{DL}$</i>	5.61	9.22	10.53	1.02	4.42
<i>Ours w/o Bypass Network</i>	4.73	7.04	10.36	1.00	4.48
<i>Our full model</i>	3.07	3.91	6.95	0.41	2.03

Table 4. Ablation study on our Fashion-body dataset. We report the variations of chest size, waist size, and hip size between the predicted circumference and the ground truth circumference.

4.5. Ablation Study

We investigate the impact of $\mathcal{L}_{DL}$ and bypass network on final body measurement accuracy. As shown in Tab. 4, the design of bypass network incorporating the $\mathcal{L}_{DL}$ loss is highly effective. This is attributed to concatenating additional features obtained from bypass network to the backbone network. Fig. 1 illustrates that our focused network structure enhances the feature extraction capability of the SMPLer-X module through residual connections. Consequently, the final extracted features are significantly enriched. The original network structure focuses on learning the global body

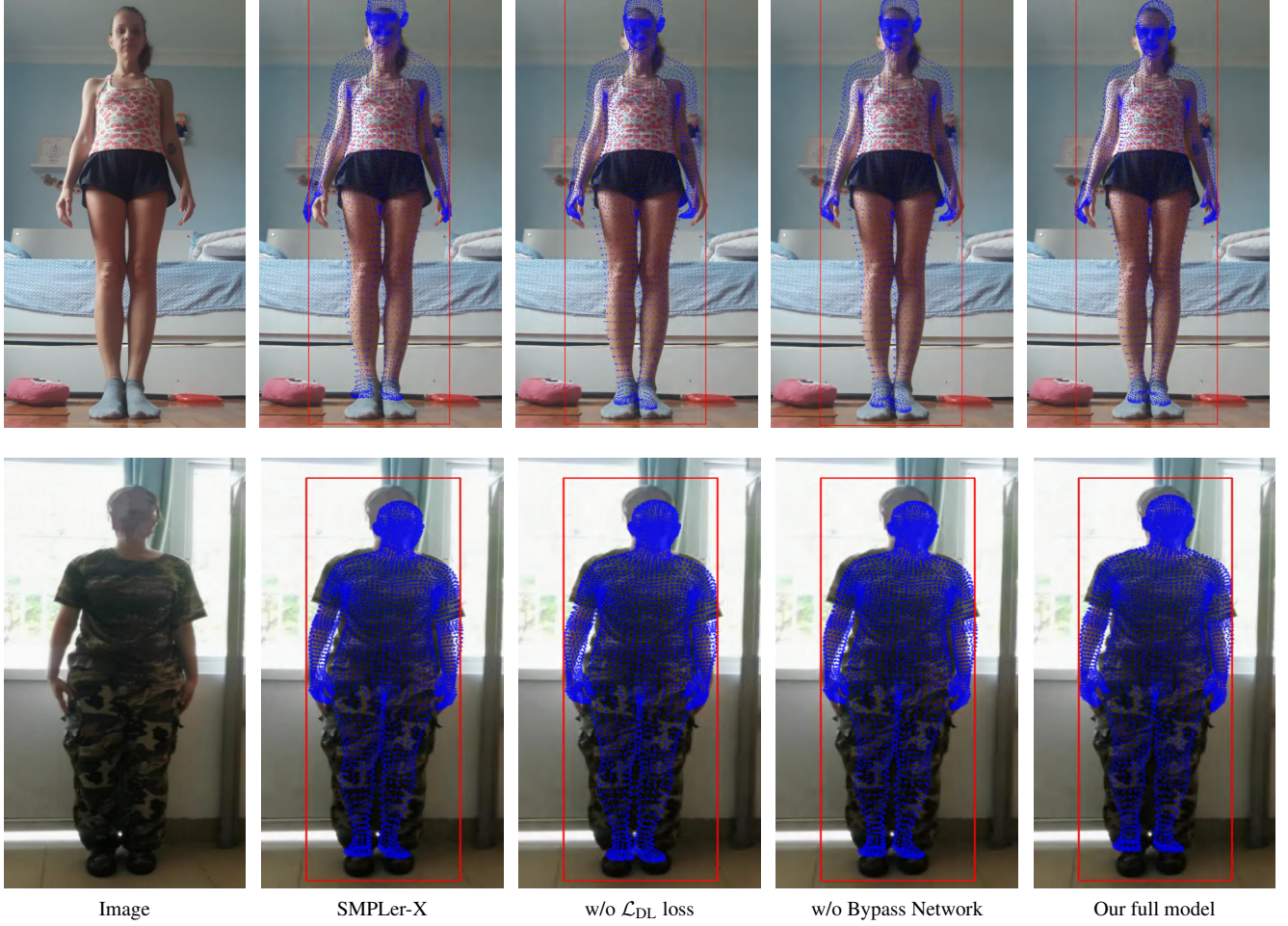


Figure 6. Ablation result of our proposed loss function and bypass network on synthetic and realistic scenes. Our full model performs best in terms of 3D key points and depth maps. From left to right, we compare the experiment result with the original SMPLer-X, w/o bypass network and ours. Both bypass network and $\mathcal{L}_{DL}$ improve the final results.

information, which is unnecessary in our measurement task. The $\mathcal{L}_{DL}$ loss term directs bypass network focus on specific measurement parts, enabling the production of accurate measurements in the subsequent stage. As shown in Tab. 4, without the addition of the $\mathcal{L}_{DL}$ loss, the predicted measurement values are less accurate. The best results are achieved when both the $\mathcal{L}_{DL}$ loss and the bypass network are integrated into our framework. As shown in Fig.6, we perform the ablation study for the Bypass Network and w/o $\mathcal{L}_{DL}$ loss on two examples from the open-source dataset and our Fashion-body dataset. We also investigate how the accuracy of different reconstruction modules affects the final body measurement error. Our full model performs best in terms of 3D key points and overall measurement parts. From left to right, we compare the experiment result with the original SMPLer-X, w/o $\mathcal{L}_{DL}$, w/o bypass network and our measurement approach. Both bypass network and $\mathcal{L}_{DL}$ improves the final results.

5. Conclusion

In this paper, we address the growing demand for accurate 3D anthropometric measurements in online garment shopping, aiming to streamline the production of well-fitted clothing. Existing methods relying on 3D scans encounter challenges in data collection and generalization across diverse scenarios. To overcome these challenges, we introduce a novel approach featuring a scaling-up human body model (SMPLer-X) capable of extracting measurements from both images and lidar data. A dynamic loss function and a Bypass Network based on CNN and ResNet architectures are integrated to enhance the effectiveness of feature extraction. And a dataset called Fashion-body is constructed to support model training and evaluation as well as future research on anthropometric measurements. Experimental results on both existing and proposed datasets demonstrate that our method is superior to current baselines and commercial software, offering a promising direction to cost-effective and tailored clothing production.

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